

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 24 Hemorrhagic Shock

Dr. Gabrielle Walcott Bedeau

Email: gwalcott@sgu.edu

Zoom: <https://sgu.zoom.us/my/meetwithdrwalcottbedeau>

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Assigned Reading

- Rhoades, R. & Bell, D. (2012); *Medical Physiology: Principles for Clinical Medicine*, 4th Edition; Wolters Kluwer. Chapter 17

Clinical cases:

Adapted from Cases and Problems in Physiology

Objectives adapted from: The American Physiological Society-Medical Curriculum Objectives(2011),
Cardiovascular

Practice Questions: BRS, Guyton, Pretest

Lecture Objectives

Lecture Objective	
SOM.1a.BPM1.3.CPR .2.PHYS.0183	Describe the direct cardiovascular consequences of the loss of 30% of the circulating blood volume on cardiac output, central venous pressure, and arterial pressure.
SOM.1a.BPM1.3.CPR .2.PHYS.0184	Describe the compensatory mechanisms activated by these changes. Explain three positive feedback mechanisms activated during severe haemorrhage that may lead to circulatory collapse and death.
SOM.1a.BPM1.3.CPR .2.PHYS.0185	Contrast the change in plasma electrolytes, hematocrit, proteins, and oncotic pressure following resuscitation from haemorrhage using a) water, b) 0.9% NaCl, c) plasma, and d) whole blood

Lecture Objective	Lecture Outline (case study)
SOM.1a.BPM1.3.CPR .2.PHYS.0183	<i>Types of shock?</i> Determine the type based on the effect on CO, SVR and PCWP
SOM.1a.BPM1.3.CPR .2.PHYS.0184	<i>What are the compensatory mechanisms?</i> <i>Baroreceptor, RAAS, Internal transfusion</i> <i>What happens when the body no longer compensates? Decompensated</i>
SOM.1a.BPM1.3.CPR .2.PHYS.0185	<i>What is the management of shock?</i> <i>Review of fluid therapy</i>

What is Shock (circulatory)?

Emergency room as seen on tv

Life-threatening disorder; Inadequate organ perfusion, tissue hypoxia, metabolic disturbance, organ damage.

There are TWO main categories of shock:

- Low stroke volume/Hypovolemic ($\downarrow$ CO)
- Vasodilation/ Distributive ($\downarrow$ TPR)

Clinical Physiological Features:

Vital signs:

- $\uparrow$ Respiratory rate: rapid, shallow
- $\uparrow$ Heart rate (tachycardia): rapid, weak, small stroke vol
- $\downarrow$ Blood pressure: Systolic <90 mmHg; Diastolic <60 mmHg
- $\downarrow$ Pulse Pressure: although (MAP may be normal)
 - $\uparrow$ sympathetic causes cold, sweaty, clammy/diaphoretic skin (hypovolemia) or warm and flushed skin (distributive)*

End-organ hypotension:

- Kidney: $\downarrow$ Urine Output: Oliguria <30 ml/hr
- Brain: Altered consciousness (drowsiness), agitation/irritability, confusion/disorientation, lethargy, stupor, weakness, collapse

Clinical Case Problem: Hypovolemic Shock: Regulation of Blood Pressure

-adapted from Cases and Problems in Physiology, Costanzo

- Mavis Bryn is a 78-year-old widow who was brought to the Emergency department one evening by her sister. Early in the day, Mrs. Bryne had seen bright red blood in her stool, which she attributed to hemorrhoids. She continued with her daily activities: she cleaned her house in the morning, had lunch with friends, and volunteered in the afternoon as a “hugger” in the newborn intensive care unit. However, the bleeding continued all day, and by dinner time, she could no longer ignore it. Mrs. Bryne does not smoke or drink alcoholic beverages. She takes aspirin, as needed, for arthritis, sometimes up to 10 tablets daily.
- In the emergency room, Mrs. Bryne was lightheaded, pale, cold and very anxious. Her hematocrit was 29% (normal for women, 36 to 46%). **What are the pertinent positives?**

What are the pertinent positives?

- 78-year-old widow **Demographics**
- **lightheaded, pale, cold and very anxious.** **Chief complaint**
- bright red blood in her stool, which she attributed to hemorrhoids. **Hx of presenting illness**
- the **bleeding continued all day**, and by dinner time, she could no longer ignore it.
- does not smoke or drink alcoholic beverages. **Social history**
- takes aspirin, as needed, for arthritis, sometimes up to **10** tablets daily. **Medications**
- hematocrit was 29%, (normal for women, 36 to 46%)
- **Labs/Objective findings**
- Can you make a diagnosis or is more information? **Vital signs**

What type of shock did Mrs. Bryne have?

Mrs. Bryne's Blood pressure and Heart rate:

Parameter	Lying down (supine)	Upright (standing)
Blood pressure	90/60	75/45
MAP	70	55
Heart rate	105 beats/ min	135 beats/min

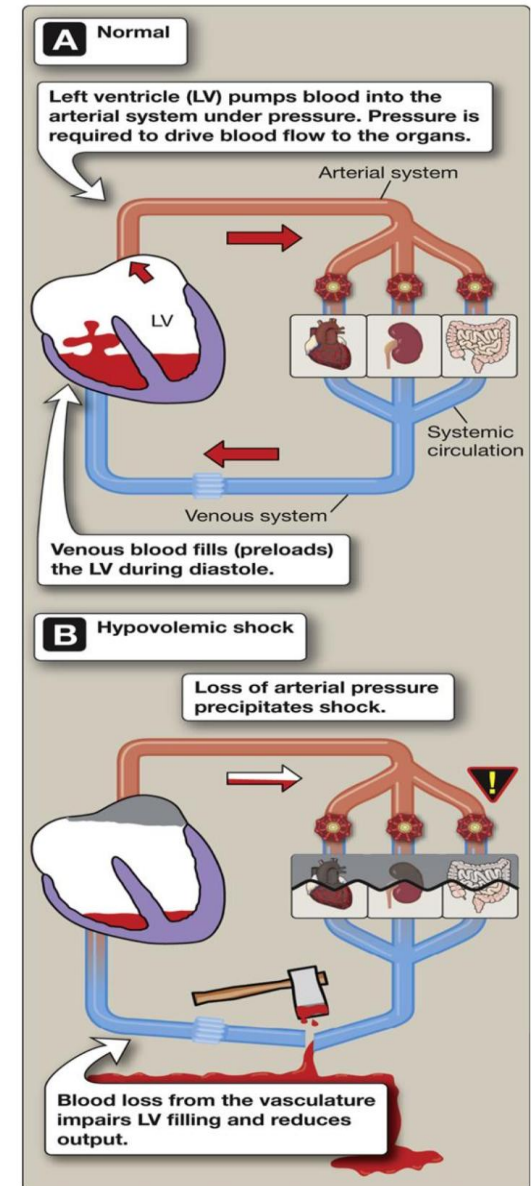
Hypovolemic shock

Hypovolemic shock is caused by a decrease in circulating blood volume. When blood volume falls, the extent to which the left ventricle (LV) fills during diastole (i.e., LV preload) decreases, which compromises CO.

When CO falls, mean arterial pressure (MAP) falls also, which reduces the amount of oxygenated blood reaching tissues.

Hypovolemic shock can be further divided into two categories:

- hemorrhagic shock and
- shock caused by loss of extracellular fluid (ECF).



Let us first discuss the progression of Hypovolemic Shock

– Low stroke volume/hypovolemia

Etiology:

- Fluid loss: (GI tract, skin, kidneys, third spacing)
- Hemorrhage

Hemodynamic changes:

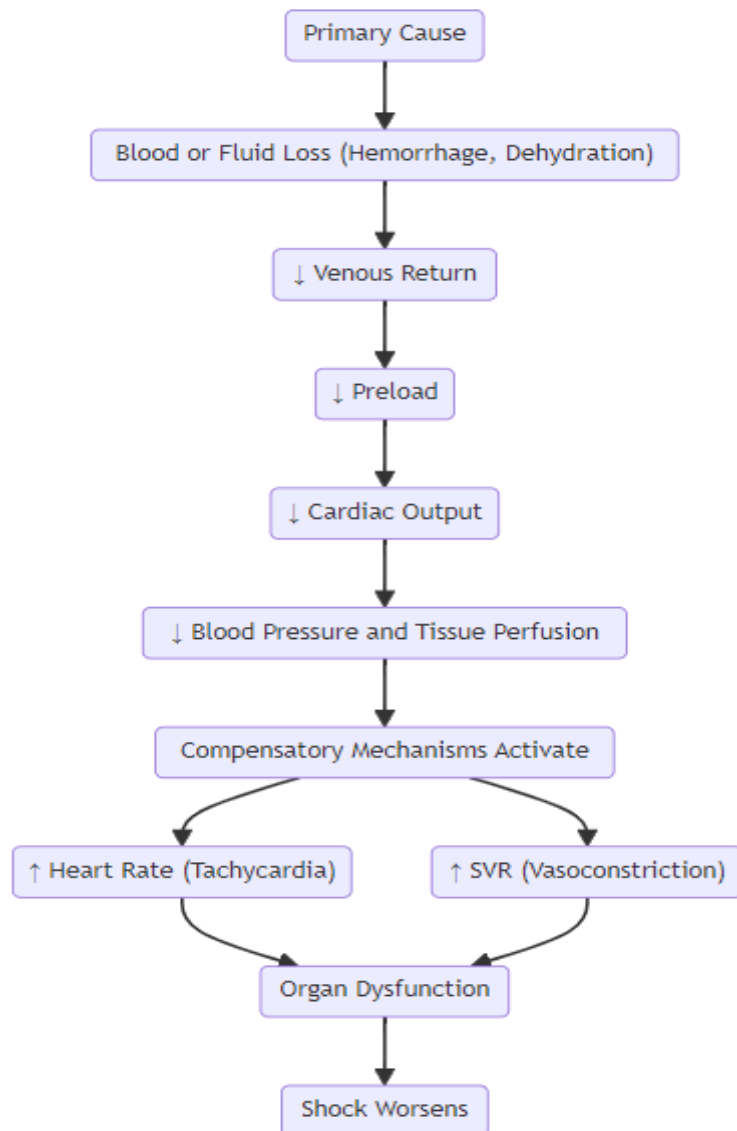
- $\downarrow$ CVP, $\downarrow$ PCWP, $\downarrow$ CO, $\uparrow$ SVR, $\uparrow$ HR, $\downarrow$ SVO₂

Clinical features:

- **Cold, clammy** extremities, slow capillary refill
- Nondistended jugular veins (sometimes referred to as “empty heart”)
- Decreased skin turgor, dry mucous membranes
- Features of underlying etiology: e.g., signs of GI bleeding, diarrhea

Treatment:

- Fluid resuscitation
- Blood transfusion
- Hemostatic control



Working your way through the clinical presentation

Shock

- Low volume vs distributive?

Cold, clammy extremities

- Peripheral temperature?

Low cardiac output

- MAP? Increased or decreased?

Jugular Venous Pressure

- PCWP – is there back up?

Hypovolemic shock

- low volume – 2^o GI bleed

HYPOVOLEMIC SHOCK (e.g. due to Hemorrhage)

What type of shock did Mrs. Bryne have?

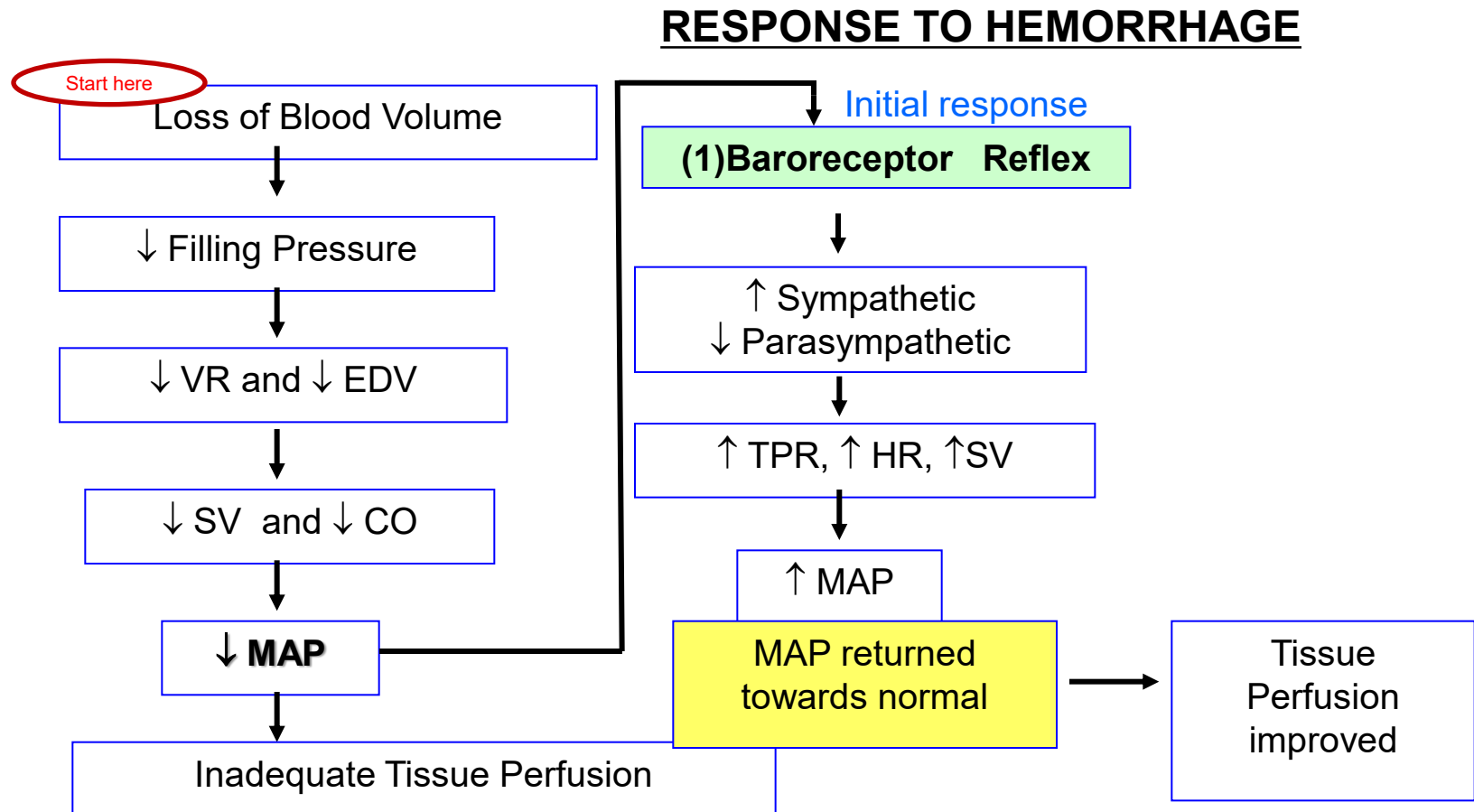
Normal Blood Volume: ~ 5 L (~65% in veins)

Consequences of Blood Loss

5 – 10%	(blood loss) No change in MAP * *Baroreceptor compensates
15 – 20%	Modest ↓ MAP: reflexes → Full Recovery
20 – 30%	MAP 60 – 80 mm Hg; rarely fatal
30 – 40%	MAP 50 – 70 mm Hg ; serious shock responses which could become irreversible
> 50%	Fatal

After the gastrointestinal blood loss, what sequence of events led to Mrs. Bryne's **decreased arterial pressure**?

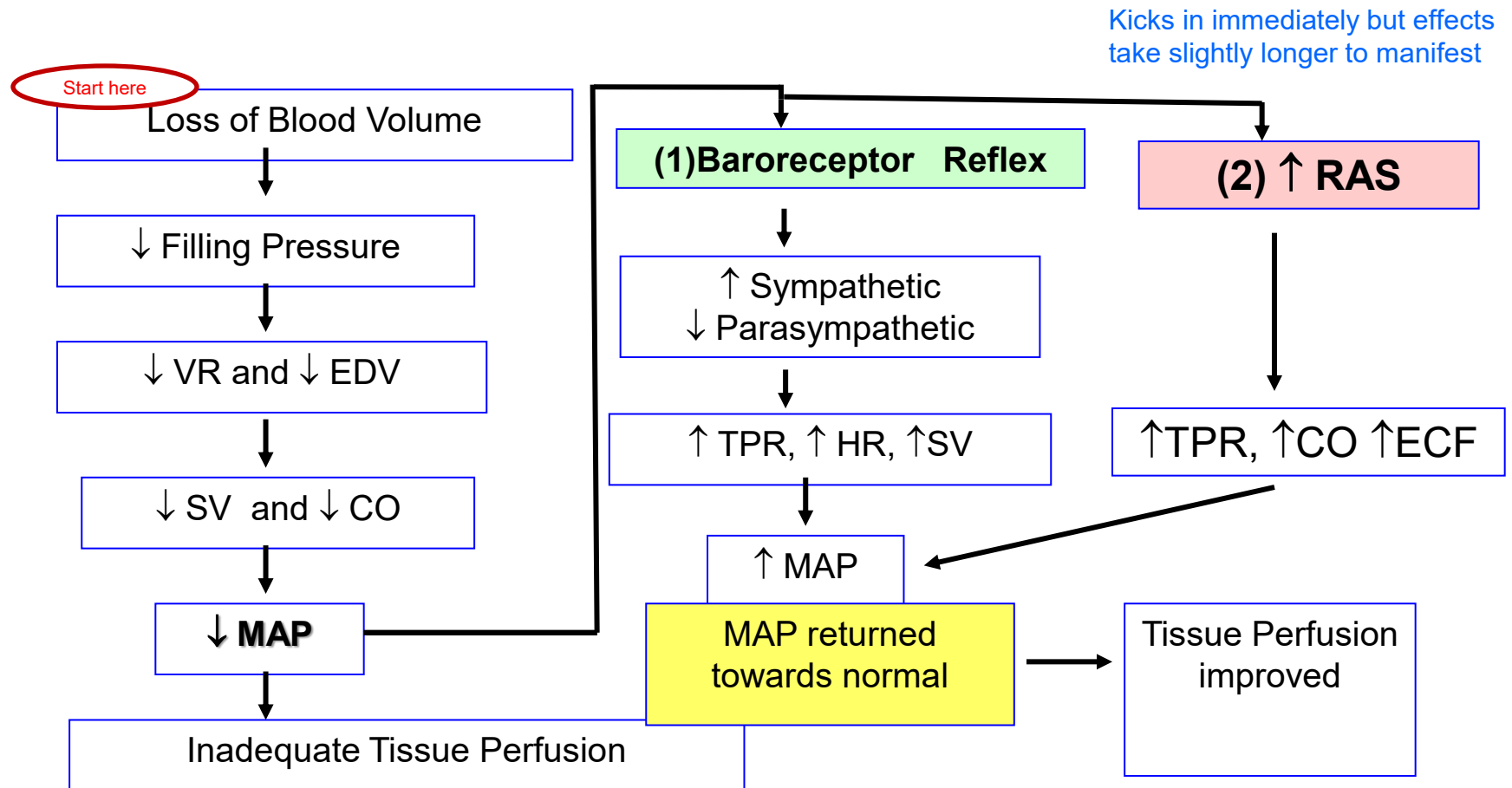
Short term responses (occurs within seconds): : **1. Baroreceptor reflex** - to preserve perfusion of heart and brain



After the gastrointestinal blood loss, what sequence of events led to Mrs. Bryne's **decreased arterial pressure**?

Long term (kicks in immediately – effects more prolonged: **2. Activation of the Renin – Angiotensin System**: to replace fluid loss

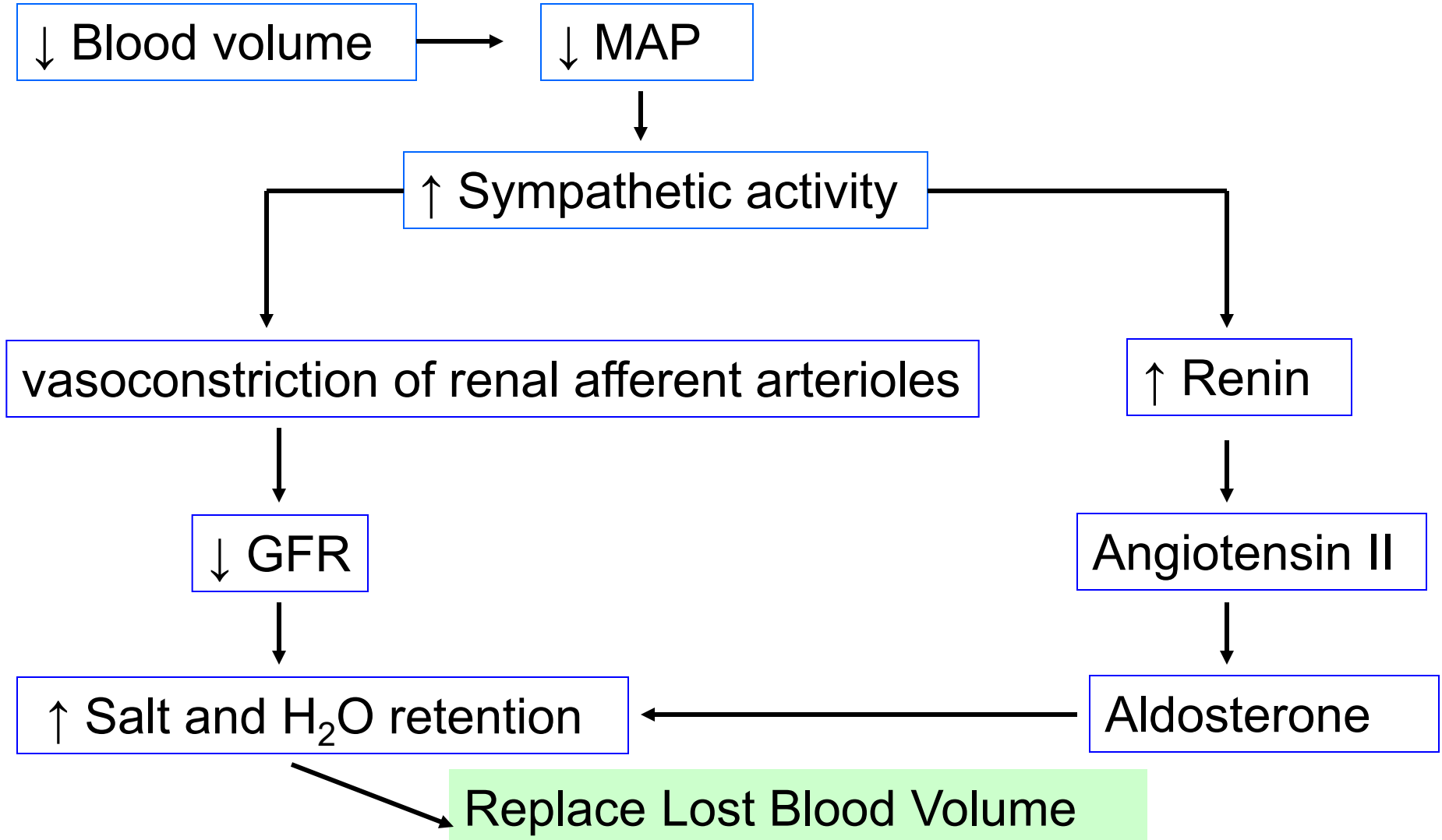
RESPONSE TO HEMORRHAGE



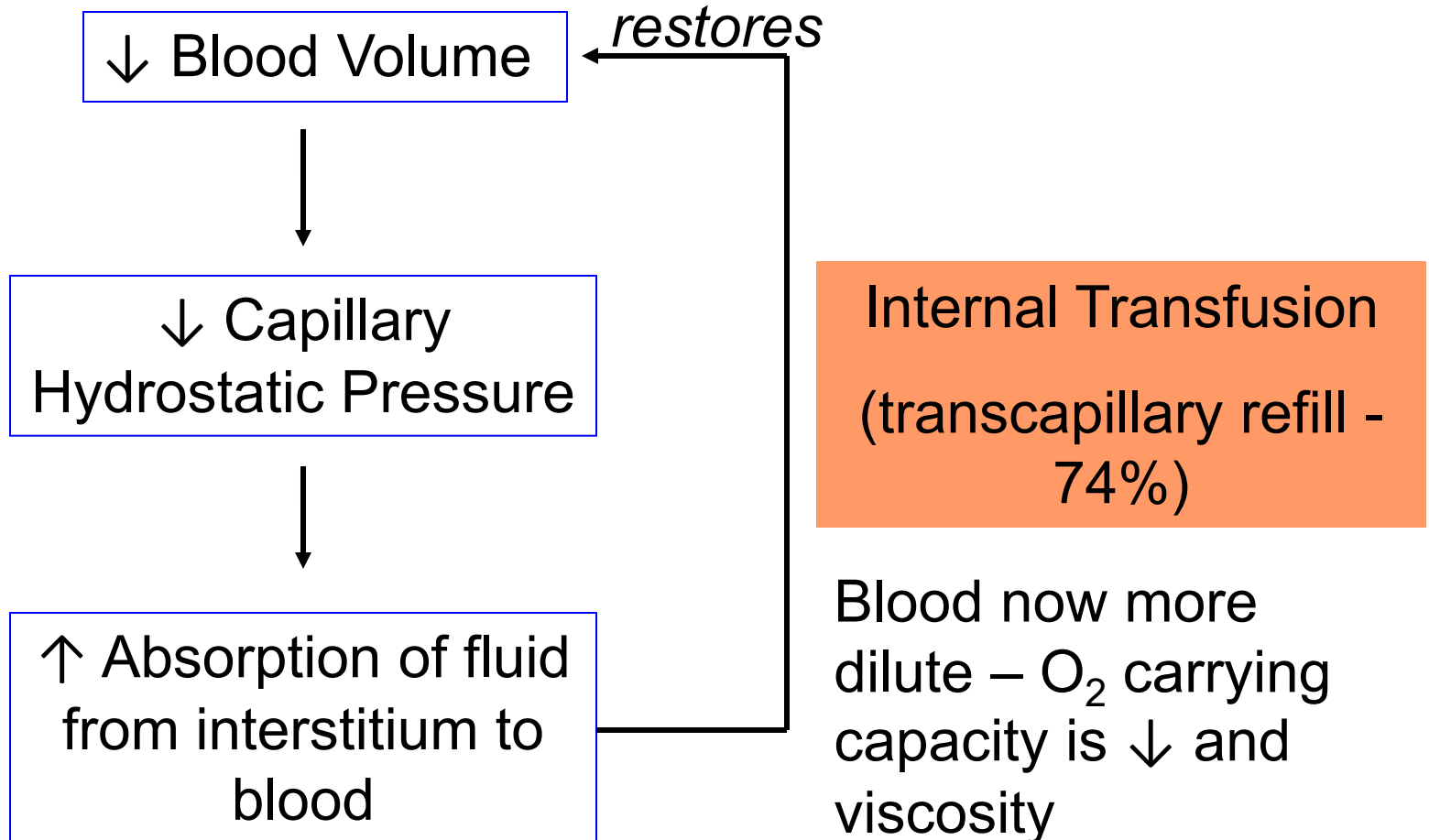
The Renin Angiotensin System - PLUS

(2) ↑ RAS

Kicks in immediately but effects take slightly longer to manifest



Another Consequence of Hemorrhage



Other Early Responses to Haemorrhage

1. ↓ Stimulation of volume receptors in atria



↓ ANP production → acts on renal tubules



Salt and water retention

2. ↑ Plasma proteins by liver (~ 7 days)

3. ↑ Erythropoietin → ↑ RBC production (4 – 6 wks)

Atrial Natriuretic Peptide released from myocytes in response to stretch. Acts on receptors to ↑ cGMP.

What happens when the body no longer compensates?

Severe, Prolonged Shock: Decompensated Shock

Severe, untreated prolonged shock leads to decompensated shock (fatal).

Occurs: Loss of $> 30\%$ blood volume, no fluid replacement for 3 – 4 hr

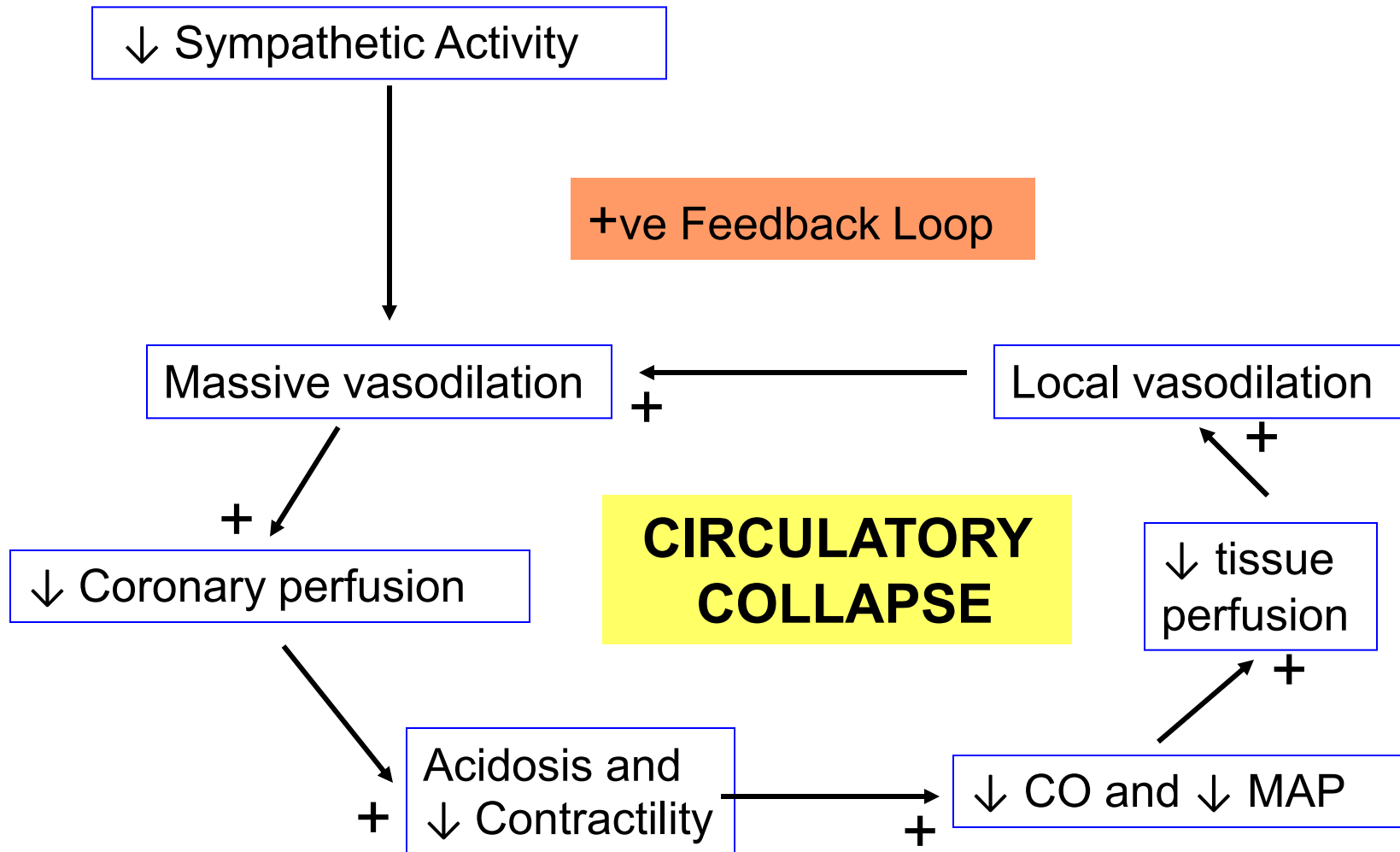
Initially: $\uparrow$ sympathetic activity (BP may be maintained)

Later: If the tissue hypoxia is not relieved, the hypoxic tissues release massive amounts of metabolic vasodilators (in an attempt to increase organ blood flow).

These metabolic vasodilators overrides the beneficial effect of sympathetic mediated- vasoconstriction. $\downarrow$ sympathetic activity (“sympathetic escape”)

Deleterious Positive feedback loops occur

Mechanism of Decompensated Shock



Summary

Decompensated Shock

Severe Hemorrhage

↓ CO

Positive
Feedback
Loops!
(2)

↓ Tissue blood flow

↓ Contractility

↓ MAP

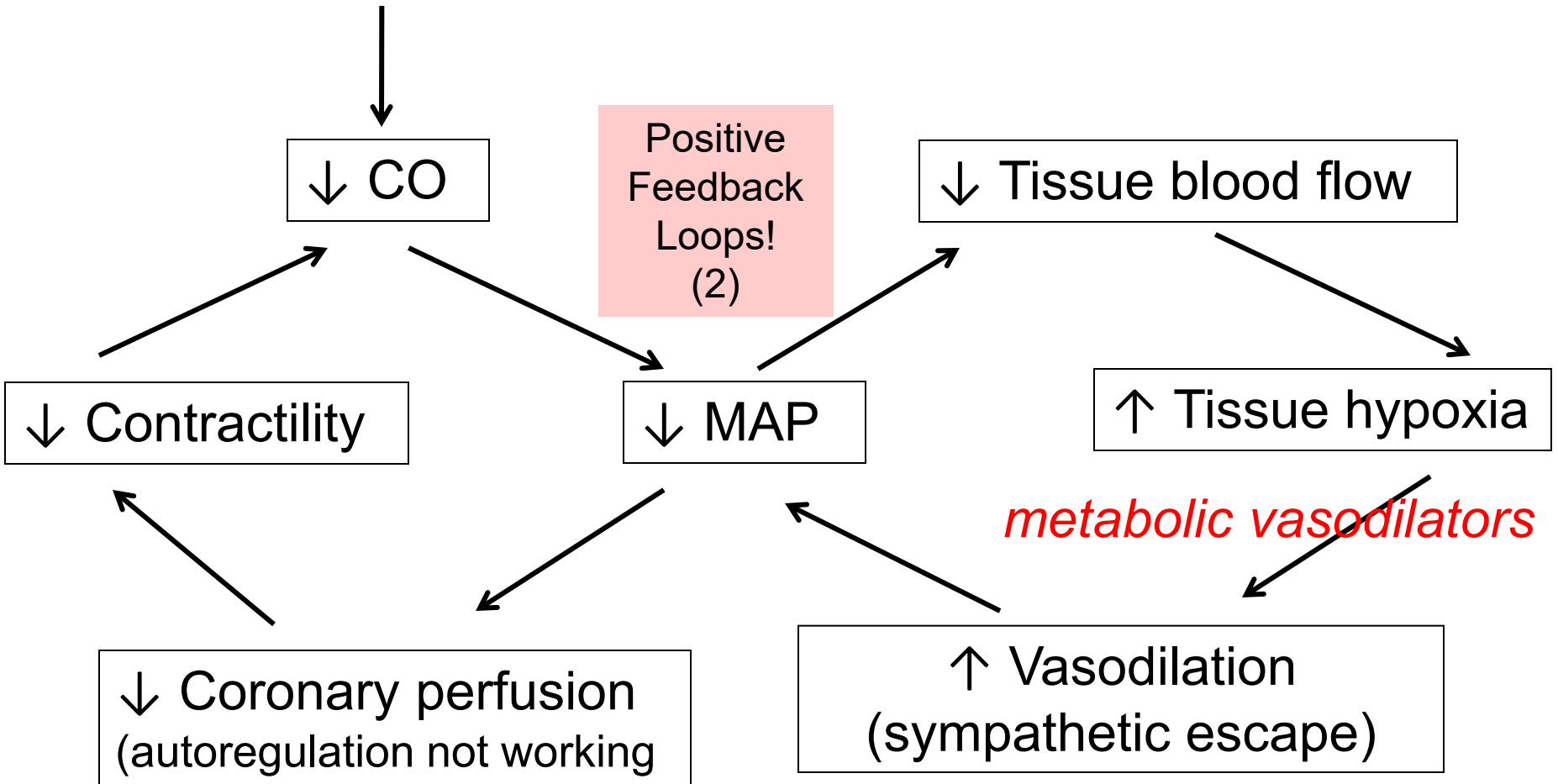
↑ Tissue hypoxia

↓ Coronary perfusion
(autoregulation not working)

↑ Vasodilation
(sympathetic escape)

metabolic vasodilators

Massive tissue vasodilation overrides the increased sympathetic vasoconstriction (“sympathetic escape”)



Other +ve Feedback Loops in Decompensated Shock

Tissue hypoxia due to ↓ perfusion



Metabolic acidosis



↓ contractility



↓ CO and TPR



↓ MAP even more

Cerebral ischemia



Initially strong
Symp activation



Vasomotor centre of the
medulla become hypoxic



↓ Sympathetic drive

What would blood lactate levels be?

Management of Shock

- Treat the underlying cause if possible
- Restore blood volume by infusion of fluids and where appropriate blood
- Ensure adequate lung ventilation and provide extra oxygen.
- Use interventions that improve cardiac performance

Mrs. Bryne's treatment

- An infusion of normal saline was started and a blood sample was drawn to be typed and cross matched to prepare for a blood transfusion.
- A colonoscopy showed that the bleeding came from herniations in the colonic wall, called diverticula. By the time of the colonoscopy the bleeding had stopped spontaneously.
- Because of the quantity of blood loss, Mrs. Byrne received two units of whole blood and was admitted for observation. The physicians were prepared to insert a bladder catheter to monitor the urine output.
- The next morning, her normal color had returned, she was no longer light-headed, and her blood pressure, both lying and standing had returned to normal.
- She was discharged and advised to take it easy...

Resuscitation

- Using a 16 gauge IV. Catheter infuse Normal saline (NaCl) at a rate that will keep systolic B.P. >100 mmHg. and pulse < 100/ min
- Transfuse:
 1. haemodynamic instability despite Normal saline infusion
 2. Hb < 7g/dl in low risk patients
 3. Hb < 9g\dl in high risk patients (elderly, coronary artery disease)

Give fresh frozen plasma for coagulopathy

Platelets for platelet count < 50,000 or platelet dysfunction

- Give supplemental oxygen
- Consider ventilation if there is - Intractable hypoxaemia, hypercapnia $P_a\text{CO}_2 > 50$ mmHg, Respiratory distress, Impaired conscious level

Monitor airway, clinical status, vital signs, urine output, N/G tube output if one in place



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https://www.terumo-europe.com/sites/productcatalog/Lists/ProductImages/versatus_EN.jpg?RenditionID=9

Therapeutic options to maximise cardiac function

- **Reduce afterload** e.g. Nitrates (arteriolar dilator), counterpulsation intra-aortic balloon
- **Increase preload** e.g. giving fluids to increase filling pressures, in a setting with impaired myocardial contractility, will only produce a small increase in SV and CO.
- **Improve myocardial contractility** e.g. an inotrope, dobutamine which in most patients acts as a vasodilator
- **Control heart rate** – the optimum heart rate 90-110 beats/min. Correction of low K and Mg first step in treating tachyarrhythmias.

After Hemorrhage, resuscitation with each of the solutions would

Solution	Electrolytes	Plasma Proteins	Oncotic Pressure	Haematocrit
Water	[C]↓	[C]↓	↓	↓
0.9% saline	No change	[C]↓	↓	↓
Plasma	No change	No change	No change	↓
Whole blood	No change	No change	No change	No change

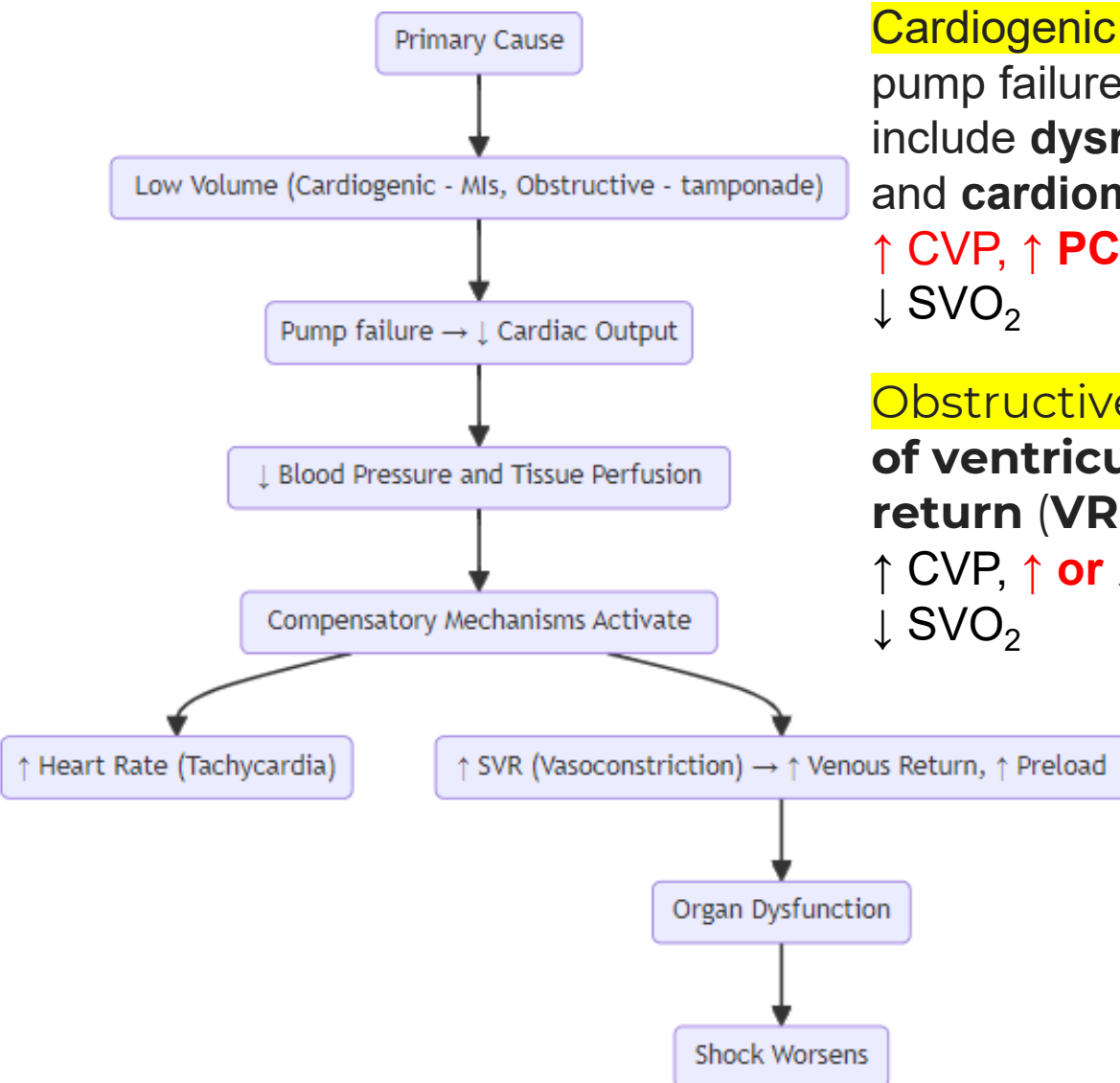
Syncope

- **Definition:** Sudden transient loss of consciousness due to a fall in MAP resulting in reduced cerebral perfusion

Mechanisms of syncope

- **Postural hypotension:** physiological peripheral vasoconstriction on standing is impaired resulting in hypotension
 - Normally a decrease in systolic BP (5 –10 mmHg), a decrease in diastolic BP and an increase in HR (10 –25 bpm). Definition of postural hypotension –a decrease in systolic BP of ≥ 20 mmHg or a decrease in diastolic BP of ≥ 10 mmHg within three minutes of standing.
- **Cardiac Syncope:** mechanical cardiac dysfunction or arrhythmia
- **Neurogenic syncope:** abnormal autonomic reflexes causing bradycardia and/or hypotension

Other types of Low Volume Shock – Cardiogenic and Obstructive



Cardiogenic shock is caused by cardiac pump failure. Common causes include **dysrhythmia**, **mechanical issues**, and **cardiomyopathies**.

↑ CVP, ↑ **PCWP**, ↓↓ CO, ↑ SVR, Variable HR, ↓ SVO₂

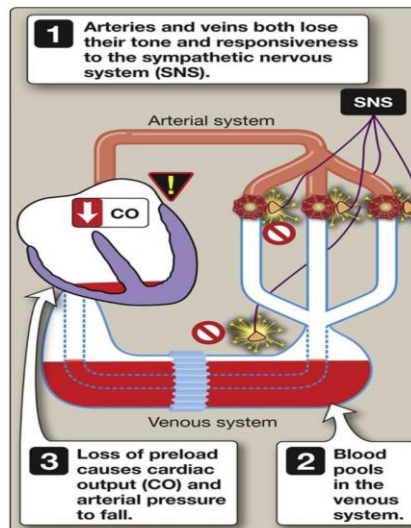
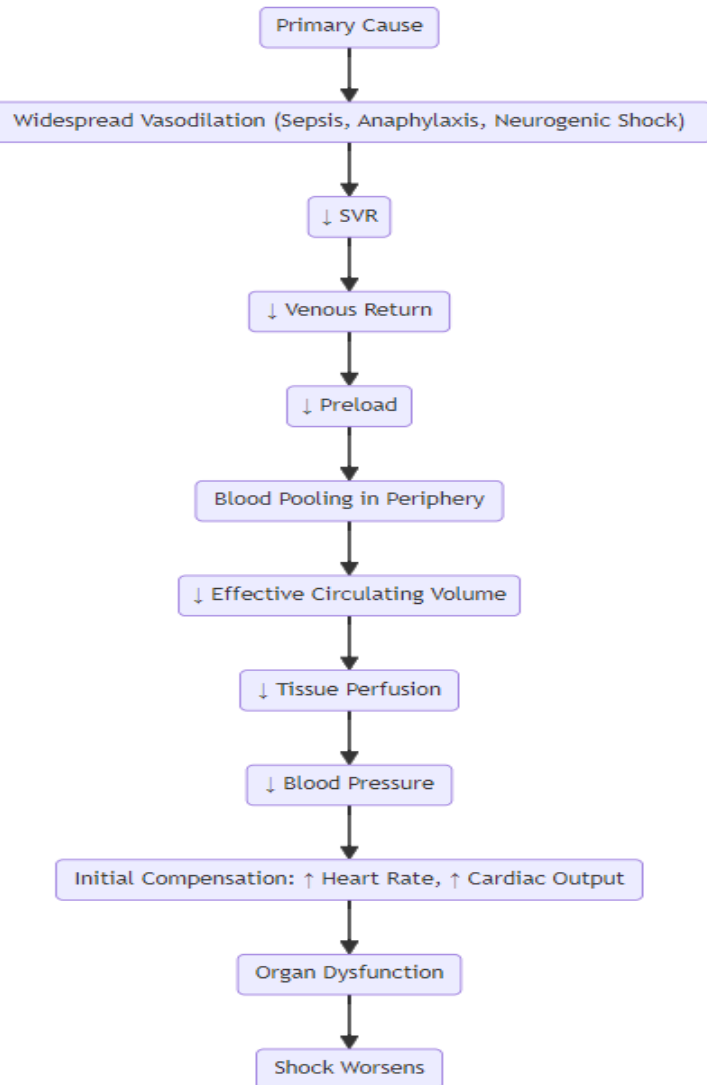
Obstructive shock is due to **obstruction of ventricular outflow** or **venous return (VR)** or **impaired diastolic filling**

↑ CVP, ↑ **or** ↓ **PCWP**, ↓↓ CO, ↑ SVR, ↑ HR, ↓ SVO₂

A quick look at Distributive Shock

Distributive Shock

Most arterial and venous vessels have a resting tone controlled by the sympathetic nervous system (SNS) as a way of limiting cardiovascular capacity to 5 L. Distributive, or **vasodilatory**, shock occurs when SNS vascular control is compromised, resulting in vasodilation and an exponential increase in vascular capacity. MAP dissipates rapidly as blood surges through dilated resistance vessels and becomes trapped in capillary beds and veins. Distributive shock has many causes. The most common include **sepsis**, **systemic inflammatory response syndrome** and **anaphylaxis**.



•Septic shock:

↓ CVP, **Normal or ↓ PCWP**, ↑ or ↓ CO, ↓↓ SVR, ↑ HR, ↑ SVO₂

Anaphylactic shock :

↓ CVP, ↓↓ PCWP, ↑ or ↓ CO, ↓↓ SVR, ↑ HR, ↑ SVO₂

•Neurogenic shock:

↓ CVP, ↓↓ PCWP, ↓ CO, ↓↓ SVR, **↓ HR**, Normal or ↑ SVO₂



YOU GOT THIS